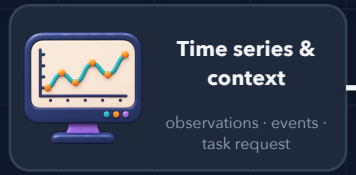
Time Series Agent: Components and Information Flow

> layers ordered from general to specific; validation feeds back to the LLM



Time series & context

observations · events · task request



LLM Side

§3 · LLM

interprets temporal evidence and selects actions

- Representation alignment
- Temporal instruction & reasoning
- Workflow policy learning

ChatTS · Time-LLM · TimeRLM

action ↓ ↑ observation



General Harness

§4 · GH

executes actions, keeps state, coordinates roles

- Interface & Interaction
- Memory & Context
- Tools & Execution
- Planning & Orchestration
- Optimization & Evolution

TimeClaw-Harness · AION · TS-Agent · TradingAgents

temporal operations ↓



General Time Series Harness

§5 · GTS

supplies temporal data, tools and time constraints

- Temporal Data Management
- Temporal Processing
- Temporal Context & Memory
- Temporal Validation

TimeSage-EV · OpenForecaster · TSQAgent · TS-Reasoner

shared temporal capabilities ↓



Task-Specific Harness

§6 · TSK

organizes outputs around task objectives

- Forecasting & Prediction
- Data Augmentation & Synthesis
- Anomaly Detection & Diagnosis
- Decision Support

FLAIRR-TS · MetaCaster · Argos · FinCon

candidate output ↓



Validation & Evaluation

§7 · EVAL

checks process, temporal validity, task and decision utility

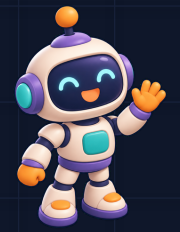
- Execution checks
- Information-cutoff checks
- Task metrics
- Downstream utility

TimeSeriesGym · Dr-CiK · TraceBench · TSFLab

general → temporal → task-specific

scope of reuse narrows downward

feedback: revise / terminate



agent loop:
act · observe · check

accept



Forecast · diagnosis · decision

accepted output